

SSH and SCP Assignment - Questions and Answers

Question 1: Use ssh to connect from your Linux system to a remote server and create a directory named 'music_downloads' in your home folder.

Answer:

Connect to Remote Server:

```
ssh user@REMOTE_SERVER_IP
```

Create Directory:

```
mkdir ~/music_downloads
```

Verify:

```
ls -ld ~/music_downloads
```

Question 2: Transfer a file named 'playlist.txt' from your local machine to the remote server's 'music_downloads' directory using scp.

Answer:

```
scp playlist.txt user@REMOTE_SERVER_IP:~/music_downloads/
```

Explanation:

playlist.txt = Local source file

user = Remote username

REMOTE_SERVER_IP = Remote server address

~/music_downloads/ = Destination directory on remote server

Verify on Remote Server:

```
ls ~/music_downloads
```

Question 3: Set up passwordless SSH login by generating an SSH key pair and copying the public key to the remote server.

Answer:

Generate Key Pair (No Passphrase):

```
ssh-keygen -t rsa -b 4096
```

Press Enter when asked for passphrase.

Copy Public Key:

```
ssh-copy-id user@REMOTE_SERVER_IP
```

Test Login:

```
ssh user@REMOTE_SERVER_IP
```

Expected Result:

Login without entering a password.

Question 4: Write a bash script that remotely restarts nginx and checks its status. Save it as remote_nginx_check.sh.

Answer:

```
#!/bin/bash
```

```
ssh user@REMOTE_SERVER_IP "sudo systemctl restart nginx && sudo systemctl status nginx --no-pager"
```

Save File:

```
nano remote_nginx_check.sh
```

Make Executable:

```
chmod +x remote_nginx_check.sh
```

Run Script:

```
./remote_nginx_check.sh
```

Question 5: Find and fix the error in this command:

```
ssh user@192.168.1.100:/home/user/music_downloads
```

Answer:

Incorrect Command:

```
ssh user@192.168.1.100:/home/user/music_downloads
```

Correct Command:

```
ssh user@192.168.1.100
```

Reason:

The ssh command is used only to connect to a remote server. Paths such as /home/user/music_downloads are used with commands like scp or after logging in to the remote system. The colon and path are invalid in a normal ssh connection command.